# Lecture 20, Math 441

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# Method of Variation of Parameters

$y'' + p(x)\,y' + q(x)\,y = f(x)$ (linear inhomogeneous)

Two linearly independent solutions to the homogeneous equation.

$y_1 : y_1'' + p\,y_1' + q\,y_1 = 0$

$y_2 : y_2'' + p\,y_2' + q\,y_2 = 0$

The complementary solution is $y_c = c_1\,y_1 + c_2\,y_2$

With the method of variation of parameters we write:

$$y_p = u_1(x)\, y_1 + u_2(x)\, y_2$$

$$y_p' = u_1'\, y_1 + u_1\, y_1' + u_2'\, y_2 + u_2\, y_2'$$

Extra condition: $u_1'\, y_1 + u_2'\, y_2 = 0$

$$y_p' = u_1\, y_1' + u_2\, y_2'$$

$$y_p'' = u_1'\, y_1' + u_1\, y_1'' + u_2'\, y_2' + u_2\, y_2''$$

$$u_1' y_1' + \cancel{u_1 y_1''} + u_2' y_2' + \cancel{u_2 y_2''} + \cancel{p u_1 y_1'} + \cancel{p u_2 y_2'} + \cancel{q u_1 y_1} + \cancel{q u_2 y_2} = f(x)$$

$$u_1' y_1' + u_2' y_2' = f(x)$$

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Put this equation together with the extra constraint:

$$\begin{cases} u_1' y_1' + u_2' y_2' = f \\ \\ u_1' y_1 + u_2' y_2 = 0 \end{cases}$$

$u_2' = -\frac{y_1}{y_2} u_1' \qquad y_1' u_1' - \frac{y_2' y_1}{y_2} u_1' = f$

$$u_1' = -\frac{y_2 f}{y_1 y_2' - y_1' y_2} \qquad u_2' = \frac{y_1 f}{y_1 y_2' - y_1' y_2}$$

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$$u_1(x) = -\int^x \frac{y_2(s)\, f(s)}{W(y_1, y_2)(s)} \mathrm{d}s \qquad u_2(x) = \int^x \frac{y_1(s)\, f(s)}{W(y_1, y_2)(s)} \mathrm{d}s$$

$$y_p = u_1(x)\, y_1(x) + u_2(x)\, y_2(x)$$

$$y(x) = y_c + y_p = c_1\, y_1 + c_2\, y_2 + u_1\, y_1 + u_2\, y_2$$

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Ex: $y'' - 4\,y' + 4\,y = 2\,\mathrm{e}^{2x}$

$y_c:\ y'' - 4\,y' + 4\,y = 0$

$y_c = \mathrm{e}^{rx} \qquad r^2 - 4r + 4 = 0 \ \to\ (r-2)^2 = 0 \ \to\ r = 2 \quad$ (twice)

$y_1 = \mathrm{e}^{2x};\ y_2 = x\,\mathrm{e}^{2x}.$ So, $y_c = c_1\,\mathrm{e}^{2x} + c_2\,x\,\mathrm{e}^{2x}$

Method of undetermined coefficients: $y_p = A\,x^2\,\mathrm{e}^{2x}\ldots.$

$y_p' = \cdots;\ y_p'' = \cdots.$ Put in equation. Find $A$.

Method of variation of parameters: $y_p(x) = u_1\,\mathrm{e}^{2x} + u_2\,x\,\mathrm{e}^{2x}$

$$y_p' = u_1' \, \mathrm{e}^{2x} + 2 \, u_1 \, \mathrm{e}^{2x} + u_2' \, x \, \mathrm{e}^{2x} + u_2 \, \mathrm{e}^{2x} + 2 \, u_2 \, x \, \mathrm{e}^{2x}$$

$$u_1' \, \mathrm{e}^{2x} + u_2' \, x \, \mathrm{e}^{2x} = 0$$

Now $y_p' = 2 \, u_1 \, \mathrm{e}^{2x} + u_2 \, \mathrm{e}^{2x} + 2 \, u_2 \, x \, \mathrm{e}^{2x}$

$$y_p'' = 2 \, u_1' \, \mathrm{e}^{2x} + 4 \, u_1 \, \mathrm{e}^{2x} + u_2' \, \mathrm{e}^{2x} + 2 \, u_2 \, \mathrm{e}^{2x} + 2 \, u_2' \, x \, \mathrm{e}^{2x} + 2 \, u_2 \, \mathrm{e}^{2x} + 4 \, u_2 \, x \, \mathrm{e}^{2x}$$

$$2\,u_1'\,\mathrm{e}^{2x} + 4\,u_1\,\mathrm{e}^{2x} + u_2'\,\mathrm{e}^{2x} + 2\,u_2\,\mathrm{e}^{2x} + 2\,u_2'\,x\,\mathrm{e}^{2x} + 2\,u_2\,\mathrm{e}^{2x} +$$

$$4\,u_2\,x\,\mathrm{e}^{2x} - 8\,u_1\,\mathrm{e}^{2x} - 4\,u_2\,\mathrm{e}^{2x} - 8\,u_2\,x\,\mathrm{e}^{2x} + 4\,u_1\,\mathrm{e}^{2x} + 4\,u_2\,x\,\mathrm{e}^{2x} = 2\,\mathrm{e}^{2x}$$

Simplifying, we get:

$$2\,u_1'\,\mathrm{e}^{2x} + u_2'\,\mathrm{e}^{2x} + 2\,u_2'\,x\,\mathrm{e}^{2x} = 2\,\mathrm{e}^{2x}$$

Adding the last result to the extra condition, we get the system

$$\begin{cases} 2\,u_1'\,\mathrm{e}^{2x} + u_2'\,\mathrm{e}^{2x} + 2\,u_2'\,x\,\mathrm{e}^{2x} = 2\,\mathrm{e}^{2x} \\ \\ u_1'\,\mathrm{e}^{2x} + u_2'\,x\,\mathrm{e}^{2x} = 0 \end{cases}$$

which has the solutions $u_1' = -2\,x$ and $u_2' = 2$

And integrating: $u_1 = -x^2$ and $u_2 = 2\,x$

Finally $y_p = -x^2\,\mathrm{e}^{2x} + 2\,x\,x\,\mathrm{e}^{2x} = x^2\,\mathrm{e}^{2x}$

and $\boxed{y = y_c + y_p = c_1\,\mathrm{e}^{2x} + c_2\,x\,\mathrm{e}^{2x} + x^2\,\mathrm{e}^{2x}}$

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